SUDARSHAN SHORNA KUMAR

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SUMMARY

Aerospace Engineering and Business Analytics student who has designed, manufactured and flight-tested [X] flight-critical rocket subsystems, secured $[amount] in industry sponsorship, and led teams of up to [X] students — seeking to bring end-to-end hardware delivery experience to industry.

EDUCATION

Bachelor of Aerospace Engineering (Honours) | Monash University, Clayton, VIC 2023 – 2027
Bachelor of Commerce (Business Analytics) | Monash University, Clayton, VIC 2023 – 2027
WAM: 76 (combined, both degrees)
IC281 Engineering Industry Skills Course | Chisholm Institute of TAFE, Dandenong, VIC — 2025
Awards: **[award / scholarship name] — [year]**

PROFESSIONAL EXPERIENCE

Flight Systems Engineer / Vice Lead | *Monash High Powered Rocketry* May 2024 – Present

- Lead a team of **[number]** engineers across recovery, propulsion and structures sub-teams, coordinating design, integration and test schedules to deliver flight-ready rockets across **[number]** launch campaigns per year.
- Designed, manufactured, integrated and flight-tested flight-critical mechanical recovery and active apogee control systems, applying DFM and DFA principles to cut subsystem assembly time by **[X]**% and part count by **[X]**%.
- Modelled systems in Creo Parametric, managed revisions in Windchill PLM, and validated designs through FEA (ANSYS Mechanical) and CFD (ANSYS Fluent) across **[number]** subsystems; presented at **[number]** Design Reviews and Flight Readiness Reviews.
- Created AS1100-compliant engineering drawings with GD&T for CNC and manual machining, supporting **[number]**+ precision machined components; contributed to 2nd-place finishes at the 2025 IREC and AURC among **[number]** competing university teams.

Business Lead | *Monash High Powered Rocketry* Mar 2023 – Jun 2024

- Led a 14-member sub-team across sponsorship, marketing, logistics and outreach, securing $**[amount]** in cash and in-kind sponsorship from industry partners including Fleet Space Technologies and Moog Australia to fund competition travel and manufacturing costs.
- Built and managed sponsor relationships with Fleet Space Technologies and Moog Australia, coordinating deliverables such as site visits, technical presentations and rocket branding to secure repeat, multi-year sponsorship.
- Grew the team's outreach program, engaging **[number]** students across **[number]** schools and university open days through hands-on STEM demonstrations and rocket-launch showcases, raising the team's profile across the Australian aerospace sector.

Retail Salesperson | *JB Hi-Fi Australia* Sept 2023 – Present

- Ranked in the top **[X]**% of sales staff store-wide by consistently exceeding revenue and product-attachment KPIs, while balancing full-time study and team leadership commitments.

TECHNICAL PROJECTS

Big Unilateral Sear Spring System (BUSSS) – Mechanical Recovery System | *Creo, ANSYS Mechanical, Windchill*

- Led development of MHPR's first fully mechanical, non-pyrotechnic parachute deployment system, replacing black-powder charges to improve ground-handling safety and enable repeatable, low-risk ground testing.
- Engineered a linear-actuator-triggered sear and spring release mechanism that vents a CO_2-pressurised piston to separate the airframe and deploy parachutes, sized through iterative FEA to reliably overcome **[X]** N of retention force before release.
- Integrated the mechanism's actuation trigger with the team's SRAD flight computer, enabling automatic deployment at apogee via barometric altitude detection, with a manual override retained for ground testing.
- Validated performance through iterative prototyping, structural FEA of the sear/spring interface, and ground testing, culminating in a successful flight deployment at 10,000 ft AGL.
- Eliminated pyrotechnic consumable and range-safety costs, cutting recovery system cost by $**[amount]** and mass by **[X]** g versus the previous system.

Air Brakes | *CAD, ANSYS Fluent, MATLAB*

- Redesigned the air brakes sled for design-for-manufacture and design-for-assembly, repositioning wire harness pass-through holes to shorten cable runs and cut sled assembly time by **[X]**%.
- Selected **[ASA / PETG]** filament for 3D-printed structural brackets to improve **[layer adhesion / impact resistance]** under flight loads while keeping production cost and lead time low.
- Redesigned the sled's battery bay to securely mount an 18650 2S battery pack, improving vibration resistance and enabling tool-free battery swaps during flightline operations.
- Designed, optimised and validated the active air brake control system using CFD modelling and custom MATLAB trajectory analysis, providing up to 20% of active apogee control.

Sled Platform with Integrated Electronics (SPINE) – Avionics Bay

- Designed and manufactured an ASA-printed avionics bay integrating PCB mounting, cable management and flight computer interfaces, reducing avionics assembly and inspection time by **[X]**% across **[number]** rocket airframes.

SKILLS

Engineering & Analysis: FEA (ANSYS Mechanical), CFD (ANSYS Fluent), Systems Engineering, Design Validation
CAD & PLM: Creo Parametric, SolidWorks, Windchill
Manufacturing: CNC & manual machining, 3D printing, Composite layup, Laser cutting, Engineering Drawings (AS1100, GD&T)
Programming & Data: Python, MATLAB, R, SQL, Data Visualisation, Machine Learning, Git